

SENTINEL-CPS

Thesis Scope Statement

Securing Cyber Physical Systems in a Remote IoT Lab: A Zero-Trust Edge Gateway Architecture Utilizing eBPF and AI Threat Detection

Project Name: Sentinel-CPS

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Primary Scope	Zero-Trust remote CPS education, TV-as-a-Track execution, and eBPF/AI threat mitigation

Prepared for Master's Thesis Development

Document Control

This scope statement defines the thesis boundaries for the current Sentinel-CPS implementation direction. It preserves the approved thesis title while clarifying the v10.0 scope around a remote educational sandbox, a hardened Zero-Trust Gateway, a dynamic TV-as-a-Track physical substrate, and planned host-level eBPF/AI threat monitoring.

Version	Date	Prepared By	Summary of Update
10.0	May 29, 2026	Tarique Chowdhury	Updated the thesis scope to reflect the Sentinel-CPS v10.0 architecture: USB-storage-based Gateway execution, RDP-accessed Bastion Host ingress, Port 8080 unified C2 application, Smart TV dynamic track generation, light-sensor PID vehicle navigation, and eBPF/AI-based active mitigation.

1. Project Objectives and Boundaries

The scope of this thesis encompasses the design, deployment, and security validation of a Zero-Trust gateway architecture for remote educational Internet of Things (IoT) and Cyber-Physical Systems (CPS) laboratories. The project establishes a strict boundary between an untrusted student-facing sandbox and a controlled physical execution environment, ensuring that remote learning activities do not expose the broader university enterprise network or the autonomous edge devices to uncontrolled access.

The thesis focuses on Sentinel-CPS as an architectural proof of concept rather than as a general-purpose robotics platform. Its primary goal is to demonstrate that remote CPS experimentation can be mediated through a hardened Gateway, auditable command path, and active threat detection pipeline while still enabling meaningful student interaction with physical IoT hardware.

- **Primary objective:** Design and validate a secure remote CPS lab pattern that combines restricted educational access, Zero-Trust gateway enforcement, dynamic physical test generation, and host-level observability.
- **Security boundary:** Keep student-facing development workflows, Gateway control services, encrypted edge telemetry, and physical actuation separated by explicit trust boundaries.
- **Experimental boundary:** Limit validation to the Sentinel-CPS prototype environment, including Raspberry Pi Gateway services, ESP32-based edge vehicles, Smart TV track execution, and simulated or controlled attack scenarios.
- **Research boundary:** Evaluate the feasibility and security value of eBPF/AI-assisted monitoring at the Gateway chokepoint rather than attempting to replace enterprise network security infrastructure.

2. In-Scope Deliverables

The following deliverables are included within the thesis scope and represent the core technical implementation and evaluation targets.

Deliverable	Scope Definition	Expected Outcome
Zero-Trust Gateway Infrastructure	Configure a bare-metal Raspberry Pi Gateway using a default-deny nftables policy, mDNS local resolution through avahi-daemon, Ed25519 SSH authentication, and high-IOPS USB storage for resilient operation under kernel logging workloads.	A hardened Gateway that acts as the central enforcement and observability node for the remote CPS lab.
Remote Educational Sandbox	Integrate a Windows-based Bastion Host accessed by RDP through the university VPN. The Bastion Host serves as the sanctioned ingress point for remote students and supports interaction with physically isolated microcontrollers.	A controlled educational access model that permits student learning without exposing the Gateway or edge network directly.
Unified C2 Web Application	Develop a localized Python Flask application hosted natively through systemd on TCP Port 8080. The application includes a Lane Builder for dynamic track generation and a synchronized Digital Twin telemetry dashboard.	A single command-and-control interface that links remote user intent, digital track design, and physical endpoint telemetry.
Dynamic Physical Substrate	Implement autonomous edge navigation using a Smart TV as a light-emitting physical floor. Custom ESP32 chassis use 12-bit analog light sensor arrays and onboard PID logic to follow digitally generated tracks.	A repeatable, reconfigurable TV-as-a-Track experiment environment that eliminates manual track construction and supports rapid CPS testing.
Active Threat Mitigation	Develop eBPF C-based kernel hooks targeting /dev/ttyUSB0 read/write syscalls and integrate telemetry with a localized AI anomaly detection model capable of triggering automated hardware-level connection severance.	A Gateway-side detection and response path capable of identifying abnormal command or telemetry behavior and initiating safe STOP/severance behavior.

3. Out-of-Scope Elements and Exclusions

To preserve experimental validity, reduce unnecessary implementation risk, and maintain a thesis-appropriate project timeline, the following elements are explicitly excluded from this research.

Excluded Element	Scope Exclusion	Rationale
Visual/Camera Payloads	Camera-based vehicle navigation, high-bandwidth HTTP video streaming, and visual computer-vision payloads are excluded from the current scope.	These payloads introduce latency, bandwidth pressure, spoofing complexity, and unnecessary attack surface compared with direct analog light-sensor navigation.
Enterprise Network Engineering	Modification, administration, or redesign of the university's 802.1X authentication, switching, routing, or institutional firewall infrastructure is excluded.	The thesis evaluates a local Gateway-controlled architecture and relies only on approved local routing/access exceptions rather than institutional network redesign.
Physical Track Construction	Static tape tracks, wood barriers, mechanical guide rails, and other manually constructed physical navigation courses are deprecated.	The TV-as-a-Track model is the selected experimental substrate because it supports dynamic, repeatable, software-defined physical course generation.
Complex Machine Learning Generation	Heavy deep-learning vision models, generative AI models, and large-scale model training are excluded from the thesis implementation.	The AI component is restricted to lightweight behavioral anomaly detection based on baseline PID telemetry, serial behavior, and expected command/telemetry patterns.

4. Assumptions and Operating Constraints

- **Approved title preservation:** All documentation and final thesis materials will retain the approved thesis title exactly as listed on the title page of this document.
- **Remote access assumption:** Remote educational access is mediated through the university VPN and Windows Bastion Host rather than by exposing the Gateway directly to the public internet.
- **Gateway constraint:** The Sentinel Gateway operates natively on bare metal through systemd, not in Docker or another containerized abstraction layer, to preserve host-level eBPF visibility.
- **Display endpoint constraint:** The Smart TV is treated as an authorized display and physical substrate endpoint, not as an administrative workstation.
- **Threat model constraint:** Attack validation focuses on controlled injection, replay, abnormal serial activity, and DoS-like behaviors relevant to the Gateway and edge telemetry path.

5. Scope Validation Criteria

The thesis scope will be considered successfully validated if the prototype can demonstrate the following outcomes under controlled lab conditions.

1. **Gateway enforcement:** The Gateway enforces default-deny access control, key-based administrative access, and service-level segmentation for SSH and the C2 web interface.
2. **Sandbox separation:** Remote student workflows remain concentrated within the Bastion Host and isolated development hardware rather than bypassing the Sentinel Gateway control boundary.
3. **Dynamic track operation:** The Lane Builder can generate a track on the Smart TV and the ESP32 edge vehicle can respond to that physical light pattern using analog sensing and PID control.
4. **Telemetry visibility:** The system produces usable Gateway logs, serial activity traces, and baseline behavioral data suitable for anomaly detection.
5. **Mitigation behavior:** When abnormal command or telemetry behavior is detected, the Gateway can issue a STOP action and initiate controlled serial communication severance to protect the physical system.

6. Final Scope Summary

This thesis is scoped as a security-focused CPS architecture project, not merely a robotics build. The central research value lies in demonstrating how a remote educational IoT lab can be made safer through Zero-Trust service exposure, hardened gateway mediation, dynamic physical test generation, kernel-level observability, and lightweight AI-assisted anomaly response.

The completed project should provide a practical and defensible reference architecture for universities seeking to teach remote CPS and IoT security concepts without granting students uncontrolled access to enterprise infrastructure or unmanaged physical devices.